

Stage 1: Warm up SFT & Verifier Construction

 **Qwen**
Pretrained VLM

 **DriveLMM-o1**
CoT Data

Supervised Fine-Tuning

Input: Image, Question
Output: <think> ...</think>
<answer> ...</answer>

GRPO
Format & Accuracy Rewards

Inference

Hard Negatives

Ground Truth

 **Qwen 235B**

Assign Max Score
Generate Scores & Critique



Curated Dataset

Input: Image, Question
model's response
Output: Score, Critique

Supervised Fine-Tuning

 **Verifier**

Stage 2: Critique-Driven Multi-Turn RL

 
Image Question

 **Qwen**
Finetuned VLM


Responses

Update Prompt



Critique

Feedback



Verifier (Frozen)

Pass & Max Turns



Composite Reward & Penalty



Format Reward



Accuracy Reward



Process Verifier Reward



Multi-Turn Penalty

Final Responses

GRPO



Stage 3: Latent Thought Distillation

Distillation Dataset

Input: Image, Question
Output: <answer> ...</answer>



Qwen
Teacher (Frozen)



<think>

...

Critique

</think>

h_T

Ground Truth

Qwen
Student (Trainable)



w/o CoT

<answer>

h_S

Output

$\mathcal{L}_{align}$

$\mathcal{L}_{CE}$

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{align}$$